

IN2901

Software Development Project

Final Proposal

A Guided MERN Stack Learning Platform & Developer Community



Group Name : DoD

Bachelor of Science Honours in Information Technology (BSc.Hons (IT))
Faculty of Information Technology
University of Moratuwa

Group Name	DoD	
Project Name	A Guided MERN Stack Learning Platform & Developer Community	
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1.Introduction

The rapid evolution of web technologies has made the MERN stack (MongoDB, Express.js, React.js, Node.js) one of the most sought after skill sets in the modern software industry. However, a significant disconnect persists between traditional academic learning and practical software engineering demands. Aspiring developers often find themselves trapped in "tutorial hell" a cycle of passive video consumption and rote code copying that fails to build genuine problem solving capabilities or architectural comprehension. Furthermore, when faced with complex production environments or standard compiler error logs, beginners frequently experience high levels of anxiety, leading to an over reliance on automated AI code generators rather than cultivating core debugging competencies.

DevCraft is conceptualized as a comprehensive, gamified learning ecosystem and developer community specifically engineered to bridge this pedagogical gap for MERN stack learners. By integrating an interactive, validation driven coding workspace, a structured theory dojo, a time attack debugging simulator, a dynamic token based AI mentorship framework, a competitive leaderboard reward system, and a peer collaboration hub, DevCraft transforms passive learners into active, resilient, and industry ready full stack engineers. The platform enforces disciplined progression, eliminates debugging fear through gamification, and establishes a sustainable bridge between junior developers and professional industry mentors.

2. Background & Motivation

In recent years, computer science education has witnessed an explosion of online learning platforms, video tutorials, and bootcamps. While these resources provide extensive theoretical knowledge, they suffer from inherent structural flaws that hinder true skill acquisition. Most platforms rely heavily on passive observation, where students mimic instructors without understanding underlying design patterns, state management complexities, or backend data flow. When these learners transition to independent project development, the absence of guided feedback loops often results in frustration and abandonment.

The motivation behind DevCraft stems from direct observations of student struggles within academic and self taught environments at the University of Moratuwa. Key pain points identified include:

- **The Illusion of Competence:** Students complete multi hour video tutorials believing they have mastered a concept, only to experience complete blockades when attempting to write code from scratch.
- **Compiler Anxiety:** Intimidating terminal error logs and stack traces trigger immediate psychological panic, preventing systematic error isolation and debugging.
- **Monotony and Burnout:** Long, unengaging coding sessions lack behavioral reinforcement, causing rapid drops in user retention and motivation.
- **Mentorship Accessibility Barriers:** While expert guidance is crucial for accelerated learning, high cost consultation models remain largely inaccessible to undergraduate and entry level developers.

DevCraft addresses these systemic bottlenecks by infusing behavioral psychology, gamification mechanics, and intelligent automated feedback into a unified MERN stack ecosystem. By turning error resolution into a rewarding challenge, incorporating real-time background code validation, and bridging AI efficiency with human expertise, DevCraft redefines how modern web development skills are acquired.

3. Problem in Brief

Despite the abundance of educational resources, current MERN stack learning methodologies fail due to three core systemic issues:

1. **Passive Dependency & Lack of Enforcement:** Conventional tutorials do not prevent copy pasting or superficial code replication. Without automated, real time background validation that locks progression until functional logic is proven, learners bypass crucial structural milestones.
2. **Psychological Barriers in Debugging:** Standard development environments treat errors as harsh blockades. Beginners lack safe, low stakes simulation spaces to practice isolating compiler bugs, syntax errors, and missing imports, creating a crippling fear of terminal logs.
3. **Fragmented Ecosystems & Retention Drop offs:** Existing tools separate theory from practice, offering isolated video players or disjointed coding grounds. Furthermore, they lack intrinsic motivation structures such as dynamic leaderboards with tangible discount incentives or anti boredom engines resulting in severe user drop off rates and lack of long term engagement.

4. Aim & Objectives

Aim

To design, develop, and evaluate DevCraft, an intelligent, gamified, and community driven MERN stack learning ecosystem that replaces passive tutorial consumption with active code engineering, structured theory mastery, systematic debugging simulations, and hybrid human AI mentorship.

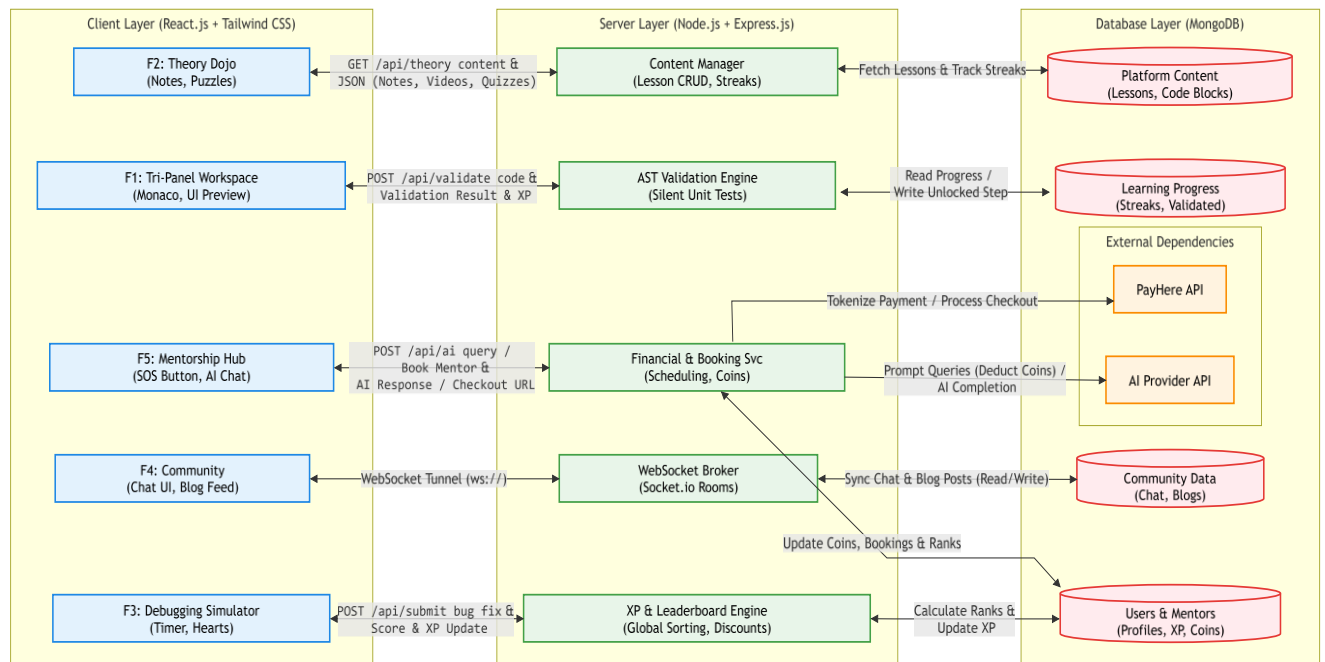
Objectives

- **To Eliminate Passive Learning via Guided Workspaces:** Implement a Tri Panel IDE featuring automated background AST (Abstract Syntax Tree) unit testing that strictly evaluates user code logic and enforces progressive step unlocking before proceeding.
- **To Cure Compiler and Debugging Anxiety:** Construct an isolated "Debugging Simulator" mini game equipped with strict countdown timers, health metrics (lives/hearts), and simulated error logs to train systematic bug isolation behavior.
- **To Ensure Comprehensive Theory Retention:** Develop "The Theory Dojo" module housing structured notes, tutorial video series, and interactive fill in the blank code puzzles reinforced by visual feedback and daily study streaks.
- **To Maximize User Retention and Engagement:** Integrate an anti boredom meme pop up engine and a competitive global leaderboard system that awards top performing users with financial discount incentives for platform services.
- **To Establish a Hybrid Mentorship & Monetization Pipeline:** Build a dual layer support system combining an AI agent (backed by a consumable coin and secure PayHere top up mechanism) with a professional human mentorship marketplace to seamlessly connect learners with verified industry experts.

5. Proposed Solution

Many undergraduate developers struggle to bridge the gap between coding theory and real world deployment, often getting stuck in "tutorial hell" or over relying on AI to solve complex errors.

DevCraft is a web based MERN stack Learning Management System (LMS) and developer community designed to break this cycle through active, hands on learning.



Architecture Diagram

Key Functionalities include:

Feature 01: The Guided Coding Workspace (Pre Defined Projects & Anti Boredom Engine)

A structured environment that transitions students from passive watching to active coding by guiding them through real MERN applications (e.g., CRUD systems).

- Core Functionalities:

- o **Tri Panel Interface:** Integrates step by step instructions, a fully functional Monaco Code Editor, and a Live UI Preview into a single, clutter free screen.

- o **Background Code Validation:** Runs silent unit tests to evaluate user code in real time. The subsequent project step is strictly locked until the user writes functionally correct code.

- o **Contextual Pop Up Checks:** Temporarily pauses the coding flow before complex logic blocks to present a mandatory theory question, ensuring conceptual understanding before syntax application.

- o **Meme Pop Up Engine:** Automatically triggers fun, context aware programming memes on the screen during long coding sessions to eliminate user boredom and maintain high engagement

Feature 02: The Theory Dojo (Notes, Video Series & Gamified Quizzes)

A comprehensive study module equipped with rich learning materials and gamified testing to master MERN stack theories.

- Core Functionalities:

- o **Structured Learning Content:** Provides normal theory notes and a tutorial lesson video series covering Beginner, Intermediate, and Advanced tiers.

- o **Interactive Puzzles:** Evaluates knowledge using "fill in the blank" code snippets rather than boring multiple choice questions.

- o **Visual & Streak Feedback:** Triggers dynamic fire animations for correct answers and maintains a "Daily Streak" counter to build consistent study habits.

Feature 03: The Debugging Simulator (Time Attack Game)

An isolated, gamified environment engineered to eliminate the anxiety caused by massive compiler errors and build bug isolation skills.

- Core Functionalities:

- o **Simulated Error Logs:** Presents users with a block of broken code alongside a realistic, intimidating error message.
- o **Time & Lives Mechanics:** Users must parse the log, locate the bug (e.g., a missing import or typo), and fix it before a strict countdown timer expires.
- o **Resilience Building:** Users are given exactly three "lives" (hearts) per session, gamifying the frustrating debugging process into an addictive challenge.

Feature 04: Dual Purpose Community (Blog, Chat & Organic Mentor Marketing)

A social knowledge sharing ecosystem designed for collaborative learning and lead generation for industry experts.

- Core Functionalities:

- o **Categorized Technical Blog:** A publishing space for users and experts to author articles segregated by specific disciplines (MERN Stack, AI, Robotics, etc.).
- o **Real Time Discussion Forum:** A WebSocket driven public chat interface allowing developers to share project milestones and crowdsource architectural advice.
- o **Organic Mentor Marketing:** Grants "Verified" badges to Mentors who post high quality solutions, building their authority and organically attracting students to book their paid sessions.

Feature 05: Hybrid Mentorship Ecosystem (Free/Coin AI Agent + Human Marketplace)

A dual layer support system providing immediate AI driven guidance alongside premium human expertise.

Core Functionalities:

- **AI Agent Mentor with Coin System:** An integrated AI assistant inside the IDE that helps debug and explain concepts. Users receive a limited stock of free coins daily; once exhausted, they can purchase additional AI query credits through secure local payments.
- **The "SOS" Button & Human Marketplace:** A panic button inside the IDE routing heavily stuck users to a professional directory of verified human mentors for 1 on 1 sessions.
- **Integrated Secure Payments:** Processes all mentor booking and coin top up transactions seamlessly via a built in local payment gateway API (e.g., PayHere).

Feature 06: Gamified Leaderboard & Discount Rewards Engine

A competitive retention engine designed to hook users to the platform through rankings and real financial incentives.

● Core Functionalities:

- **Top User Leaderboard:** Ranks active users globally based on XP earned from completed projects, quizzes, and debugging challenges.
- **Discount Reward System:** Automatically awards top ranking users on the leaderboard with exclusive financial discounts for booking human mentor sessions or purchasing platform features, driving high user addiction and continuous competition.

Suggested Technologies:

Frontend (Client Side)

- **React.js:** The core UI framework for building the single page application.
- **Monaco Editor API:** Explicitly required for the in browser IDE experience (powers the Tri Panel Interface).
- **WebContainers / CodeSandbox API:** To compile and render the "Live UI Preview" of MERN applications entirely in the browser without server roundtrips.
- **State Management (Zustand or Redux):** Crucial for tracking real time global states like the time-attack timer, user lives (hearts), XP, daily streaks, and coin balances.
- **Animation Library (Framer Motion / Lottie):** To handle the complex dynamic fire animations, meme pop ups, and gamified streak visual feedback.

Backend (Server Side)

- **Node.js & Express.js:** The core backend infrastructure to handle API routing, user authentication, and business logic.
- **Socket.io (WebSockets):** Explicitly required for the "WebSocket driven public chat interface" and real time multiplayer/forum capabilities.
- **Isolated Code Execution Engine (Docker / Pterodactyl):** A sandboxed backend environment to securely run "Background Code Validation" and evaluate malicious or infinite looping user code.
- **Automated Testing Frameworks (Jest / Mocha):** To run the silent background unit tests against the student's submitted code to unlock subsequent steps.
- **Cron Jobs (node cron):** To automate the daily resetting of free AI coins and daily streak counters.

Database & Caching

- **MongoDB:** The primary NoSQL database for storing user profiles, categorized technical blogs, quizzes, mentor availability, and transaction histories.
- **Redis (In Memory Data Store):** Highly recommended for fast retrieval of the "Gamified Leaderboard," session management, and handling real time WebSocket message brokering.

Third Party APIs & Integrations

- **Payment Gateway API (PayHere):** Explicitly required for processing local secure payments for coin top ups and mentor bookings.
- **AI Provider API (Google Gemini API or OpenAI API):** To power the "AI Agent Mentor" that assists with code debugging and concept explanations.
- **Cloud Storage (AWS S3 / Cloudinary):** For hosting high bandwidth assets like the tutorial video series, user avatars, and meme image repositories.
- **Video Hosting/Streaming Service (Mux / Vimeo API):** If building a custom video player for the "Video Series" to prevent unauthorized downloading and ensure smooth streaming.

6. Resource Requirements

Hardware & Network

- **Computer/Laptop:** Intel Core i3 / Ryzen 3 or higher, 4 GB RAM or above.
- **Storage:** At least 128 GB available storage.
- **Internet Connectivity:** Stable high speed Internet connection.

Software Requirements

- **Operating System:** Windows 10/11, Linux, or mac
- **Web Browser:** Latest version of Google Chrome, Microsoft Edge, Firefox, or Safari.
- **JavaScript Support:** JavaScript enabled modern web browser.

Additional Requirements

- **Webcam & Microphone:** Required only for supported mentor communication sessions
- **Speakers/Headphones:** Required for accessing video/audio learning content.

7. Work Distribution

1. Wijesiri K.S.K -244230C

- **Primary Responsibility:** The Theory Dojo (Notes, Video Series & Quizzes) & UI/UX Architectural Direction
- **Core Technologies:** React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Digital Graphic Design Tools (Figma/Photoshop).
- **Detailed Scope of Work:**
 - **UI/UX & Layout Architecture:** Designing the overall aesthetic, layout structuring, and responsive user experience of the entire DevCraft ecosystem.
 - **Structured Content Management:** Building the backend CRUD architecture and frontend interfaces for normal theory notes and tutorial lesson video series across Beginner, Intermediate, and Advanced tiers.
 - **Interactive Learning Puzzles:** Developing the gamified "fill in the blank" code snippet evaluation system to replace traditional boring multiple choice questions.
 - **Engagement & Retention Mechanics:** Implementing visual feedback engines (celebratory fire animations) and the "Daily Streak" tracker to encourage consistent study habits.

2. Dinisuru P.W.L-244047V

- **Primary Responsibility:** Hybrid Mentorship Ecosystem & Discount Rewards Engine
- **Core Technologies:** Node.js, Express.js, PayHere API, OpenAI/LLM API, MongoDB.
- **Detailed Scope of Work:**
 - **Coin-Based AI Agent Integration:** Developing the backend logic for the daily free coin allocation system, token consumption tracking, and integration with the OpenAI/LLM API for real time coding assistance.
 - **Secure Payment Gateway:** Implementing the local PayHere API for secure financial transactions, processing mentorship bookings and AI coin top ups.
 - **Human Mentorship Marketplace:** Constructing the professional directory, calendar availability checker, and 1 on 1 session booking system.
 - **Discount Rewards Engine:** Engineering the automated backend calculation system that links leaderboard rankings with financial discounts for platform services.

3. Madhushan M.A-244118P

- **Primary Responsibility:** The Debugging Simulator & Gamified Leaderboard
- **Core Technologies:** React.js, Tailwind CSS, Node.js, Express.js, MongoDB.
- **Detailed Scope of Work:**
 - **Time Attack Game Environment:** Constructing the isolated, gamified debugging arena featuring strict countdown timers and simulated, intimidating error logs.
 - **Resilience & Health Mechanics:** Implementing the "three lives" (hearts) system and XP reward distribution to build bug isolation confidence.
 - **Global Ranking Engine:** Developing the backend sorting algorithms and database queries to dynamically rank active users globally based on cumulative XP.

4. Kalhari L.T-244096T

- **Primary Responsibility:** Dual Purpose Community (Blog & Chat)
- **Core Technologies:** React.js, Tailwind CSS, Socket.io, Node.js, Express.js, MongoDB.
- **Detailed Scope of Work:**
 - **Technical Publishing Platform:** Building the categorized blog module allowing users and experts to author, format, and read articles across various tech domains.
 - **Real Time Discussion Forum:** Implementing WebSocket driven (Socket.io) real-time chat rooms for collaborative peer discussions and project milestone sharing.
 - **Mentor Marketing Integration:** Developing the "Verified" badge system that highlights high performing mentors based on their community contributions.

5. Gunasekara M.I.T.-244068K

- **Primary Responsibility:** The Guided Coding Workspace (Pre Defined Projects & Memes)
- **Core Technologies:** React.js, Tailwind CSS, Monaco Editor API, Jest (Unit Testing), MongoDB.
- **Detailed Scope of Work:**
 - **Tri Panel IDE Interface:** Engineering the clutter free layout integrating instructions, the Monaco Code Editor, and the Live UI Preview window.
 - **Background AST Validation Engine:** Implementing automated, silent unit testing (e.g., Jest) to evaluate user code in real time and enforce strict step progression.
 - **Contextual Pop Up Checks & Meme Engine:** Programming the logic that pauses the IDE for theory verification and triggers anti boredom programming memes during extended coding sessions.

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9. Appendix - Plan of Action

Pharse /timeline	October	November	December	January	February	March
Phase 1: req. Analysis & ui/ux						
Phase 2: core platform dev		(30%)				
Phase 3: integration & games						
Phase 4: testing & final doc						

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Supervisor Name: Ms.M.N.Chandimali

Nipuni
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(Supervisor)